Square patterns and cyclotomic degree bounds over finite fields

Human Compression project: F011 common theory

This paper reworks the character-sum and cyclotomic mechanism of Kim–Yip–Yoo [1], with explicit zero and order exclusions. Throughout, q is a power of an odd prime p, the quadratic character χ is extended by $\chi(0) = 0$, and "square" includes zero. We retain the classical multiplicative Weil estimate: if a nontrivial character has order d, and a polynomial is not a scalar multiple of a dth power and has h distinct geometric roots, its complete character sum has absolute value at most $(h-1)\sqrt{q}$. This is source Lemma 2.1, citing Lidl–Niederreiter, Theorem 5.41. It is an external theorem, not proved by this paper. The project's earlier character-sum audit used the same interface for a different application and did not recover the Diophantine-tuple results.

1 Character indicators, including their zeros

Lemma 1 (Prescribed character values)**.** *Let $d \mid q-1$, let ψ have order $d > 1$, and let $a_1, \dots, a_k$ be distinct. If ϵ_i are dth roots of unity and N counts the solutions of $\psi(x + a_i) = \epsilon_i$ for all i, then*

$$\left| N - \frac{q}{d^k} \right| \le \left(k - 1 - \frac{k}{d} + \frac{1}{d^k} \right) \sqrt{q} + \frac{k}{d}.$$

Proof. Use $P_i(x) = d^{-1} \sum_{j=0}^{d-1} \epsilon_i^{-j} \psi(x + a_i)^j$, where the zeroth term is one even at a root. Off $-a_i$, this is the indicator of the desired character value; at $-a_i$ it is $1/d$. Hence the sum of $\prod P_i$ differs from N by a number between zero and k/d. A nonconstant term in the expansion has exponents between zero and $d-1$, so its linear-factor product is not a dth power. For a support of size h the Weil bound is $(h-1)\sqrt{q}$. The total coefficient is

$$d^{-k} \sum_{h=1}^{k} \binom{k}{h} (d-1)^h (h-1) = k - 1 - k/d + d^{-k},$$

by differentiating $(1+z)^k$ and subtracting its nonconstant part. The triangle inequality proves the result. □

Lemma 2 (Weighted square filter)**.** *Let $f_1, \dots, f_T$ be nonconstant, monic, squarefree, pairwise coprime polynomials, of total degree D. Set*

$$W(x) = 2^{-T} \prod_{i=1}^{T} (1 + \chi(f_i(x))).$$

Then $0 \le W(x) \le 1$, and $W(x) > 0$ precisely when all $f_i(x)$ are squares, possibly zero. Moreover

$$\sum_{x \in \mathbb{F}_q} W(x) \ge \frac{q}{2^T} - \left(\frac{D}{2} - 1 + 2^{-T} \right) \sqrt{q}. \tag{1}$$

Proof. Every factor is zero, one or two. Every nonempty subproduct of the f_i is squarefree of positive degree, so the Weil bound applies. Each f_i occurs in 2^{T-1} nonempty subproducts. Their total degree is therefore $2^{T-1}D$, and subtracting one for each of the $2^T - 1$ subproducts gives the coefficient in (1). The filter is generally a weighted count at zeros, not the cardinality of the admissible set; positivity suffices below. □

2 Cyclotomic factors and a degree estimate

Let Φ_j denote the reduction of the integral cyclotomic polynomial. For $j = p^a b$, $p \nmid b$, $a \geq 1$, the integral cyclotomic identities and Frobenius give

$$\Phi_j(X) = \Phi_b(X)^{(p-1)p^{a-1}} \quad \text{in } \mathbb{F}_q[X]. \tag{2}$$

This is a square. If $p \nmid j$, the polynomial $X^j - 1$ is separable, so Φ_j is squarefree. Distinct prime-to-p indices give coprime factors of the separable polynomial $X^{\mathrm{lcm}(i,j)} - 1$. None of these cyclotomic polynomials vanishes at zero. These statements recover all three clauses of source Lemma 2.4.

For $m \geq 2$, put

$$I = \{1 \leq i < m : p \nmid i\}, \quad T = |I| + 1 = m - \lfloor (m-1)/p \rfloor, \quad \alpha = \frac{p}{p-1},$$

and $s = \sum_{i \in I} \varphi(i)$, $S = \sum_{i \in I} \varphi(2i)$.

Lemma 3. *The two degree sums satisfy*

$$s \leq S \leq \sum_{i \in I} i \leq K := \frac{\alpha}{2} T(T-1). \tag{3}$$

In particular $S < \alpha T^2$, as asserted in source Lemma 3.4.

Proof. For odd i, $\varphi(2i) = \varphi(i) \leq i$; for even i, $\varphi(2i) = 2\varphi(i) \leq i$. Write $m-1 = pu+v$, $0 \leq v < p$. Then $r = |I| = (p-1)u + v = T - 1$, and a sum over complete blocks gives

$$A := \sum_{i \in I} i = \frac{p(p-1)u^2}{2} + puv + \frac{v(v+1)}{2}.$$

Direct subtraction gives

$$\frac{\alpha}{2} r(r+1) - A = \frac{pu}{2} + \frac{v(v+1)}{2(p-1)} \geq 0.$$

This proves (3). It does not require Cohen's totient-sum estimate. □

Proposition 4 (Two cyclotomic square patterns)**.** *Suppose $m \geq 2$ and*

$$\sqrt{q} > \alpha \, 2^{T-2} T^2. \tag{4}$$

There is a square $y \in \mathbb{F}_q^*$ of order at least m for which every $y^i + 1$, $1 \leq i < m$, is a square. If $q \equiv 1 \pmod 4$, there is also such a y for which every $y^i - 1$, $1 \leq i < m$, is a square. These are separate choices of y.

Proof. For the plus pattern use the pairwise coprime squarefree family $X, \Phi_{2i}(X)$ for $i \in I$. Its total degree is $D = 1 + S \leq 1 + K$; the degree-one factor must be included. Apply Lemma 2. There are at most s nonzero elements of order below m, since for each possible order j there are at most $\varphi(j)$ elements, and $p \nmid j$. Each removes weight at most one. The point zero has weight $1/2$, since $\Phi_{2i}(0) = 1$. Thus the weight at eligible points is at least

$$\frac{q}{2^T} - \left(\frac{K-1}{2} + 2^{-T} \right) \sqrt{q} - K - \frac{1}{2}. \tag{5}$$

Since $T \geq 2$, we have $K + 1 \leq \alpha T^2/2$. Condition (4) gives $z := \sqrt{q} > 2^{T-1}(K+1)$. Substituting this into the first term in (5), the expression is greater than

$$(1 - 2^{-T})z - K - \tfrac{1}{2} > (2^{T-1} - \tfrac{3}{2})K + 2^{T-1} - 1 > 0.$$

Choose an eligible point of positive weight. It is nonzero and square; all $\Phi_{2i}(y)$ with $i \in I$ are squares. The polynomial identity

$$X^j + 1 = \prod_{d \mid 2j,\ d \nmid j} \Phi_d(X)$$

shows the same for $y^j + 1$: all such d are even and at most $2j$, and factors with $p \mid d$ are squares by (2). This is a polynomial identity and needs no division by a possibly vanishing $y^j - 1$.

For the minus pattern use $X, \Phi_i(X)$ for $i \in I$. Its degree is $1 + s \le 1 + K$. At zero its weight is again $1/2$ because the only negative constant term is $\Phi_1(0) = -1$, a square under the hypothesis. The same argument applies, using $X^j - 1 = \prod_{d \mid j} \Phi_d(X)$. □

Lemma 5 (Explicit parameter choice)**.** *Let $q > 7$, put*

$$B = \frac{\frac{1}{2}\log q - 2\log\log q}{\log 2}, \qquad m = \lfloor \alpha(B+1) \rfloor .$$

If $m \ge 2$, condition (4) holds.

Proof. Writing $m = pu + v$, $0 \le v < p$, directly gives $T \le m/\alpha + 1$, including the case $v = 0$. Consequently $T \le B + 2$. Set $x = \log q > 2$. Since $\log x > \log 2$,

$$T \le \frac{x/2 - 2\log x}{\log 2} + 2 < \frac{x}{2\log 2}.$$

Also $\alpha \le 3/2 < 4(\log 2)^2$. One elementary lower bound $\log 2 = \int_1^2 t^{-1} dt > 2/3$ follows from strict convexity and the midpoint rule; hence $4(\log 2)^2 > 16/9 > 3/2$. Taking logarithms, we obtain

$$\log\bigl(\alpha 2^{T-2} T^2\bigr) \le x/2 - 2\log x + \log\alpha + 2\log T < x/2,$$

which is (4). □

3 A shorter pattern with a stronger order exclusion

Proposition 6. *For an integer $n \ge 4$, define*

$$I_n = \{1 \le i \le n-2 : p \nmid i\}, \qquad T = n - 1 - \lfloor (n-2)/p \rfloor.$$

If $\sqrt{q} > \alpha 2^{T-2} T^2$, there is a square $y \in \mathbb{F}_q^*$ of order at least n such that $y^j + 1$ is a square for $1 \le j \le n-2$.

Proof. Apply Lemma 3 with $m = n - 1$, and set $K = \alpha T(T-1)/2$. The family $X, \Phi_{2i}(X)$ for $i \in I_n$ has degree at most $1 + K$. This time we exclude all orders below n, not merely those below $n - 1$. Their number is at most $s + \varphi(n-1) \le K + n - 1 \le K + \alpha T$; the last inequality follows from $T \ge (n-1)/\alpha$. Exclude zero at weight $1/2$ as before. Writing $z = \sqrt{q}$, the remaining weight is at least

$$\frac{z^2}{2^T} - \left(\frac{K-1}{2} + 2^{-T}\right) z - K - \alpha T - \frac{1}{2}. \tag{6}$$

Here $T \ge 3$, so $K \ge \alpha T$. Also $K + 1 \le \alpha T^2/2$, and the hypothesis gives $z > 2^{T-1}(K+1)$. Thus (6) is greater than

$$(2^{T-1} - \tfrac{3}{2})K - \alpha T + 2^{T-1} - 1 \ge (2^{T-1} - \tfrac{5}{2})K + 2^{T-1} - 1 > 0.$$

The same cyclotomic product identity as in Proposition 4 now gives every required power condition. The extra order exclusion is essential for the centered construction. □

Lemma 7 (Parameter choice for the stronger bound)**.** *Suppose $q > 7$ and $q \equiv 7 \pmod 8$. Put*

$$B = \frac{\frac{1}{2}\log q - 2\log\log q}{\log 2}, \qquad n = \lfloor \alpha(B+2) \rfloor.$$

If $n \geq 4$, the hypothesis of Proposition 6 holds.

Proof. The characteristic satisfies $p \equiv 7 \pmod 8$, the extension degree is odd, and $p \geq 7$. Write $n = pu + v$, $0 \leq v < p$. If $v = 0$, then $T = (p-1)u = n/\alpha$; if $v = 1$, then $T = (p-1)u + 1 = n/\alpha + 1/p$; and if $v \geq 2$, then $T = (p-1)u + v - 1 = n/\alpha + v/p - 1$. Therefore

$$T \leq n/\alpha + 1/p \leq B + 2 + 1/p. \tag{7}$$

This replaces the insufficient rounding estimate in the source proof.

Set $x = \log q$. The smallest eligible prime power beyond seven is at least 23, so $x > 3$. Since $2 + 1/p \leq 15/7 < 3$ and $9 > 8$, we have $2\log x > (2 + 1/p)\log 2$. Consequently (7) implies $T < x/(2\log 2)$. Moreover

$$\alpha 2^{1/p} \leq \tfrac{7}{6}\, 2^{1/7} < \tfrac{7}{4} < \tfrac{16}{9} < 4(\log 2)^2;$$

here $2^{1/7} < 3/2$ and the final inequality was proved in Lemma 5. Hence

$$\begin{aligned} \log(\alpha 2^{T-2} T^2) &\leq x/2 - 2\log x + \log(\alpha 2^{1/p}) + 2\log T \\ &< x/2. \end{aligned}$$

This proves the stated strict inequality. □

References

[1] S. Kim, C. H. Yip and S. Yoo, *Explicit constructions of Diophantine tuples over finite fields*, Ramanujan J. **65** (2024), 163–172, doi:10.1007/s11139-024-00888-5.

[2] R. Lidl and H. Niederreiter, *Finite Fields*, second edition, Cambridge University Press, 1997, Theorem 5.41 (retained through the exact statement in [1, Lemma 2.1]).